1. Steps towards a minimal model.

① Adjoining new var. (→ so that $\partial^2(x)=0$.)

Given a dga A, given $z \in Z_n(A)$, we adjoin $|x| = |z|+1$, $\partial(x)=z$, and
- exterior var. when $|x|$ odd (for graded comm.): $A[x] = A \oplus Ax$
- polynomial var. when $|x|$ even: $A[x] = A \oplus Ax \oplus Ax^2 \oplus \cdots$.

Then $A[x_1, \cdots, x_n]$ constructed in this way is the <u>free graded comm. alg. generated</u> by $x_1, \cdots, x_n$.

- Adjoining a var. only kills n-cycle.

$$H_i(A[x]) = \begin{cases} H_i(A), & i < n \\ H_n(A)/H_0(A)[z], & i = n. \end{cases}$$

Here we view $H_n(A)$ as $H_0(A)$-mod.

pf: The elements of degree $n+1$ are: $A_{n+1} \oplus A_0 x$, so boundaries are $\partial(A_{n+1})$ and $\partial(A_0 x) = A_0 \partial(x) = A_0 z$, and cycles are the same.

So we say cycle z is <u>killed</u>.

② Procedure to create a dga resolution for $R = Q/I$.

i) Initial step: for $I = (f_1, \cdots, f_m)$, we let $X_1 := \{x_1, \cdots, x_m\}$, and let:

$$Q[X_1] = Q[x_1, \cdots, x_n : \partial(x_i) = f_i],$$

for $|x_i| = 1$.

Since the degree is odd, $x_i x_j = -x_j x_i$, $x_i^2 = 0$, and $Q[X_1]$ is exactly

Kos(I).

ii) d-th step : assume we have constructed $Q[X_{\le d}]$. Then pick $t_1, \cdots, t_p$ of degree d s.t. $[t_1], \cdots, [t_p]$ generates $H_d(Q[X_{\le d}])$.

Then $X_d := \{s_1, \cdots, s_p\}$ s.t. $\partial(s_i) = t_i$, and:

$$\begin{cases} \text{adjoin exterior var. if } |s_i| = d+1 \text{ is odd} \\ \text{poly. othw.} \end{cases} \Rightarrow H_d(Q[X_{\le d+1}]) = 0$$

(by ① H_d is killed).

Let $X := \bigcup_d X_d$, so the res. is $Q[X]$.

Rmk : $f_1, \cdots, f_n$ reg. $\Rightarrow$ stop at step 1) ; othw. $H_1 \neq 0$, so must proceed.

③ Semifree extensions.

Def : A semifree extension is just adjoining variables to kill cycles, exterior / polynomial depending on degree.

Def : A minimal model $Q[X]$ for R over Q is a semifree extn. over Q that resolves $Q/I = \hat{R}$, where we take the smallest no. of var. each time. ☆☆

(min. reg. repn.)

Now we present some e.g. of minimal models.

i) $Q = k[x,y,z]$, $I = (xy, zx, yz)$, compute min. model for $R = Q/I$.

- $X_1 = \{T_{11}, T_{12}, T_{13}\}$ s.t. $\partial(T_{11}) = f_1 = xy, \cdots$.

☆ Note that in 1st step we view Q in deg. 0.

• Then we compute $H_1(\mathbb{Q}[X,]$, and $\begin{cases} \text{at degree } 1, \bigoplus_{i<j} \mathbb{Q}T_{1i}T_{1j} \\ \text{at degree } 0, \bigoplus_i \mathbb{Q}T_{1i} \end{cases}$

So $\begin{cases} B_1 = \partial(\bigoplus_{i<j}\mathbb{Q}T_{1i}T_{1j}) = (f_iT_{1j} - f_jT_{1i})_{i\neq j} \\ Z_1 = \{(a,b,c) : axy+bzx+cyz=0\}, \text{ so } y|b, z|a, x|c, \text{ and} \end{cases}$

it's generated by $Z_1 = (zT_{11} - yT_{12}, zT_{11} - xT_{13})$ and their difference $xT_{13} - yT_{12}$.

Note that both generators are not killed, so let $X_2 = \{T_{2,1}, T_{2,2}\}$, where $\partial(T_{2,1}) = zT_{11} - yT_{12}$, $\partial(T_{2,2}) = zT_{11} - xT_{13}$.

So the first two steps are $\mathbb{Q}[X_1, X_2]$.

ii) $Q = k[[x,y]]$, $I = (x^2, xy)$, and $R = Q/I$.

• $X_1 = \{T_{1,1}, T_{1,2}\}$, $\partial(T_{11}) = f_1 = x^2$, $\partial(T_{12}) = f_2 = xy$.

• B_1 is $xyT_{12} - x^2T_{12}$, Z_1 is $(yT_{11} - xT_{12})$.

Then $X_2 = \{T_{2,1}\}$, $\partial(T_{21}) = yT_{11} - xT_{12}$.

• Degree 2: $\mathbb{Q}T_{11}T_{12} \oplus \mathbb{Q}T_{21}$,

$\partial(aT_{11}T_{12} + bT_{21}) = ax^2T_{12} - axyT_{11} + byT_{11} - bxT_{12}$

$\Rightarrow \begin{cases} by - axy = 0 \Rightarrow ax = b \Rightarrow a = 1, b = x. \\ ax^2 - bx = 0 \end{cases}$

Thus $X_3 = \{T_{31}\}$, $\partial(T_{31}) = T_{11}T_{12} + xT_{21}$

④ Properties of minimal models as <u>min. dga</u>.

i) Let $\mathbb{Q}[X]$ be a min. model for R over $\mathbb{Q}$.

- $\partial(X_1) \subset m^2$.

  $\partial(X_1) = I \subset m^2$

- $\partial(X_2) \subset mX_1$

✡✡ Note that $\partial(X_n)$ must span $H_n(\mathbb{Q}[X_{<n}])$ since Z_{n-1} is killed in this way (the case $H_{n-1} = 0$ is trivial).

Now $\partial(X_2) = \sum_{x \in X_1} a_x x$ since in degree 1 there's only $\mathbb{Q}X_1$.

Then $\sum_{x \in X_1} a_x [\partial(x)]$ spans $H_1(\mathbb{Q})$, and by minimality $a_x \in m$ (otherw. there's unit & $\exists$ a relation, or we can say $[\partial(x)]$ is not L.I. in H_1/mH_1).

Thus $\partial(X_2) \subset mX_1$

- $\partial(X_{n+1}) \subset mX_n$ might fail for general n.

See 1.3.11. in BFJ.

Rmk ✡✡: i) is just a corollary of ii): the first two degree doesn't have quadratic terms, while the others might have.

ii) ✡ Differentials of minimal model.

Thm: Let $\mathbb{Q}[X]$ be a minimal model for R with max. $(X_{\geq 0})$ (where X_0 gen. m), then:

$$\partial_{\mathbb{Q}[X]}(X) \subset (X)^2.$$

$k[X] = \mathbb{Q}[X] \otimes_{\mathbb{Q}} k$ with max. $(X_{\geq 1})$. Then:

$$\partial_{k[X]}(X_{\geq 1}) \subset (X_{\geq 1})^2.$$

pf: (more than 2 var.)

- We write $\partial(y)$, $y \in X_{d+1}$ in quadratic + linear terms.

$$\partial(y) = w + \sum_{x \in X_d} a_x x$$

$w \in (X_{\geq 1})^2$, $a_x \in Q$

- Applying ∂, $\quad 0 = \partial(w) + \sum_{x \in X_d} a_x \partial(x) \in Z_d(Q[X_{\leq d}])$.

✡ Here $w \in Q[X_{\leq d}]$ since it's like yz for $y, z \in X_{\leq d}$, so $\partial(w)$ is a boundary in $Q[X_{\leq d}]$ (only linear terms span H_d)

Thus $0 = \sum_{x \in X_d} a_x [\partial(x)] \in H_d(Q[X_{\leq d}])$.

- $\{[\partial(x)] : x \in X_d\}$ is a minimal gen. set of H_d, so by NAK, $a_x \in \mathfrak{m}$.

$$\Rightarrow \quad \partial(y) = w + \sum_{x \in X_d} a_x x \in (X_{\geq 1})^2 + \mathfrak{m}(X_{\geq 1}) \subset (X_{\geq 0})^2.$$

Def: Let A be dga s.t. $A_0 = (R, \mathfrak{m})$. A is minimal dga if for $J = \mathfrak{m} + (A_{\geq 1})$ the max. ideal,

$$\partial(J) \subset J^2.$$

Note it's far from requiring a min. res., which needs $\partial(F) \subset \mathfrak{m}F$ (all coeff. in $\mathfrak{m}$, not just linear terms.)

Thus minimal models are minimal dgas. Indeed:

iii) $Q[X]$ is a semifree extn. of Q. $Q[X]$ is min. model for $R = Q/I$ over Q if and only if:

$$\begin{cases} H_n(Q[X]) = R \text{ if } n = 0 \text{; } 0 \text{ othw.} \\ \partial(X) \subset (X)^2. \end{cases}$$

pf: ✡ This is basically ii) in reverse:

- We want to prove $[\partial x]_{x\in X_{d+1}}$ is a minimal generating set for $H_d(Q[X_{\le d}])$.
  First $[\partial x]$ generates H_d since for $t\neq 0$ in $H_d(Q[X_{\le d}])$, $t\neq\partial(t')$ for $t'\in Q[X_{\le d}]$, but $H_d(Q[X])=0$, so $t\in\partial(QX_d)$.
- Assume $\sum a_x[\partial x]=0$, WTS $a_x\in m$ so $[a_x]=0$ in H_n/mH_n.
  Then $\sum a_x\partial x+\partial w=0$ for $w\in Q[X_{\le d}]$, since $\deg(w)=d+1$ we have $w\in(X)^2$.
- Then $\partial(\sum a_x x+w)=0\Rightarrow \sum a_x x+w\in H_{d+1}(Q[X])$

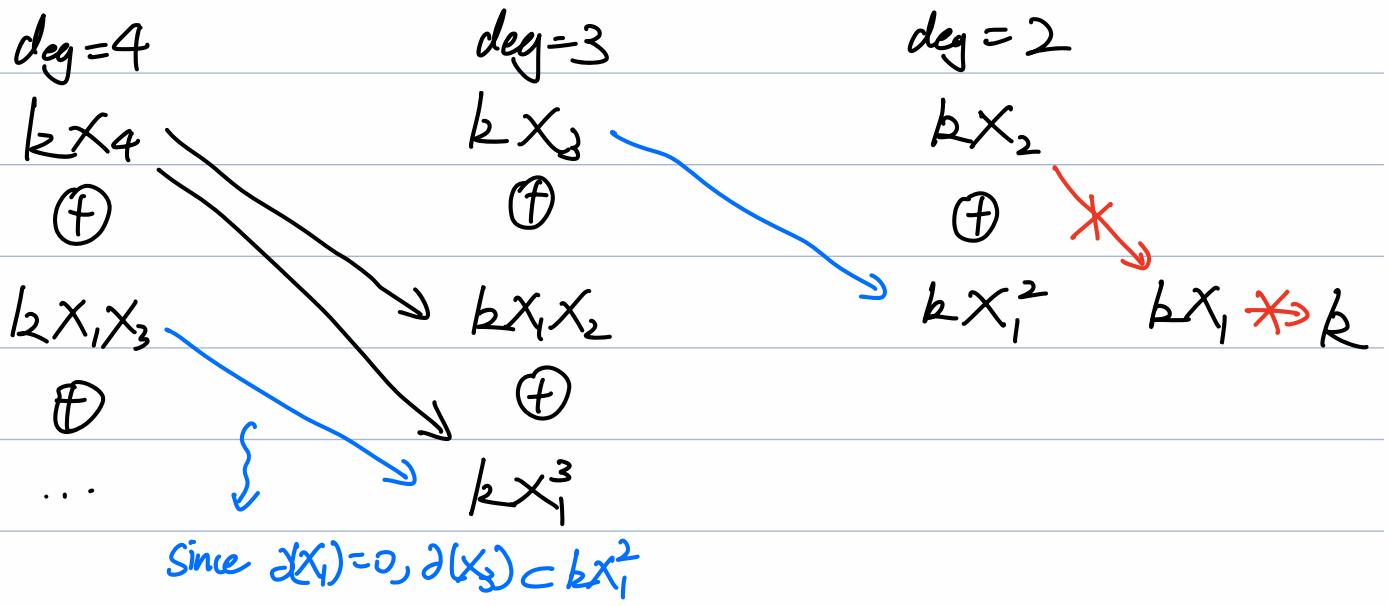
$$\xrightarrow{H_{d+1}=0}\ \sum a_x x+w=\partial(y)$$

$$\xrightarrow{\partial(X)\subset(X)^2}\ \sum a_x x+w\in(X)^2\Rightarrow \sum a_x x\in(X)^2.$$

Thus $a_x\in m$ since othw. there would be linear terms with $a_x\notin m$.

- Then $[\partial x]$ is min. gen.

iv) Then we can easily characterise the differentials of $k[X]$: eg.



2. Uniqueness properties of semifree & minimal model.

Semifree extensions:

① $A \to B$ induces unique $A[X] \to B$.

Thm: $A[X_z : \partial(X_z) = z] \to B$ is the unique map extending:

$\varphi: A \to B$, $\tilde{\varphi}(\partial(X_z)) = \partial(\tilde{\varphi}(X_z))$.

$\tilde{\varphi}(X_z) := b_z$ s.t. $\partial(b_z) = \varphi(z)$.

② $\lambda: A \to B$ induces unique $A[X] \to B[X]$

Thm: $\lambda: A \to B$ induces:

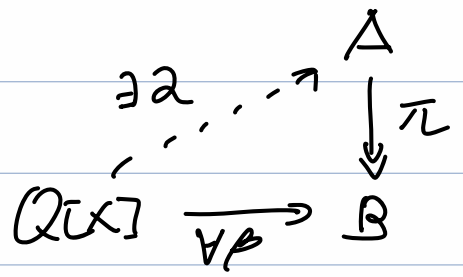
$\tilde{\lambda}: A[X_z : \partial(X_z) = z] \longrightarrow B[X_z : \partial(X_z) = \lambda(z)]$.

where if λ is quasi-isom., surj. or local, so is $\tilde{\lambda}$.

③ Lifting property of $Q[X]$ free res.

Thm: Let $Q[X]$ be a semifree extn. that is also a free res of Q/I.

Then:



& λ is unique up to Q-lin. homotopy.

④ ☆ Uniqueness of semifree extn.

Thm: Any two semifree extensions A & B of Q resolving Q/I

are homotopy equiv.; also, any quasi-isom. $A \to B$ is hom-t. equiv.

Minimal models:

⑤ Thm: Any two minimal models for R over Q are isomorphic dgas.

⑥ Reg seq. & minimal models.

Lem: Q/I min. reg. repn. of R. Then TFAE:

i) I is generated by reg. seq.

ii) $X = X_1$, $Q[X]$ is the Koszul complex on min. gen. set of I.

iii) $Q[X]$ for R is a finite free res. for Q/I.

iv) $Q[X]$ is min. res. for Q/I.

Pf: Let $\underline{f} = f_1, \dots, f_n \in I$ be s.t. $\alpha(X_1) = \{f_1, \dots, f_n\}$. Either way, we have $\underline{f}$ is min. gen. set for I & $Q[X_1] = Kos(\underline{f})$. Then:

$H_1(Q[X_1]) = 0 \iff \underline{f}$ is a reg. seq.